

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

PHYSICS — SSC-I (9th Grade)

Syllabus: Unit 7 — Density and Temperature

Total Marks: 65

Time Allowed: 2 hours 30 minutes

Version: 1

General Instructions:

1. This paper has **three sections**: A (MCQs), B (Short Questions), C (Long Questions).
2. **Section A** (12 marks): 12 MCQs. Mark the correct bubble. Time: 15 minutes.
3. **Section B** (33 marks): 11 short questions $\times$ 3 marks each. Answer *either* the main question *or* the OR alternative.
4. **Section C** (20 marks): 4 long questions $\times$ 5 marks each. Answer *either* the main question *or* the OR alternative.

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks) [15 min]

Q.1: Choose the correct answer and fill in the corresponding bubble.

#	Question	A	B	C	D	Answer
1	Density is defined as:	Volume per unit mass	Mass per unit volume	Weight per unit area	Force per unit mass	☐ A ☐ B ☐ C ☐ D
2	The SI unit of density is:	g/cm^3	kg/m^2	kg/m^3	g/mL	☐ A ☐ B ☐ C ☐ D
3	A cube of side 5 cm has mass 500 g. Its density is:	2 g/cm^3	4 g/cm^3	0.4 g/cm^3	10 g/cm^3	☐ A ☐ B ☐ C ☐ D
4	Osmium is the densest material at room temperature. Its density is:	$13,600 \text{ kg/m}^3$	$22,590 \text{ kg/m}^3$	$19,300 \text{ kg/m}^3$	$7,900 \text{ kg/m}^3$	☐ A ☐ B ☐ C ☐ D
5	In the displacement method, volume of the irregular solid equals:	Initial water volume	Final water volume	$V_f - V_i$	$V_i - V_f$	☐ A ☐ B ☐ C ☐ D
6	Plasma is the fourth state of matter because it consists of:	Neutral atoms only	Positive ions and free electrons	Tightly bound lattice particles	Liquid droplets at high pressure	☐ A ☐ B ☐ C ☐ D
7	Temperature is directly proportional to the average:	Potential energy of particles	Kinetic energy of particles	Mass of particles	Volume of particles	☐ A ☐ B ☐ C ☐ D
8	Absolute zero on the Celsius scale is:	0°C	-100°C	-273°C	273°C	☐ A ☐ B ☐ C ☐ D
9	Boiling point of water on the Kelvin scale is:	100 K	273 K	373 K	212 K	☐ A ☐ B ☐ C ☐ D
10	37°C converted to Fahrenheit is:	98.6°F	66.6°F	105°F	37.6°F	☐ A ☐ B ☐ C ☐ D
11	Thermometric property of a liquid-in-glass thermometer is:	Change in resistance	Change in emf	Volume expansion of liquid	Change in colour	☐ A ☐ B ☐ C ☐ D
12	A narrower capillary tube in a liquid-in-glass thermometer gives:	Wider range, less sensitivity	Greater sensitivity, faster response	Better linearity, slower response	Lower sensitivity, wider range	☐ A ☐ B ☐ C ☐ D

SECTION B — Short Answer Questions ($11 \times 3 = 33$ Marks)

Q.2: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Part	Question	Marks	OR	OR Question	Marks
(i)	Define density . Write its formula and SI unit. State whether density is a scalar or vector quantity and give a reason.	3	OR	Show that $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$. A petrol sample has density 0.9 g/cm^3 . Express this in kg/m^3 and state why liquids are commonly measured in g/mL .	3
(ii)	A wooden block: mass 300 g, volume 150 cm^3 . A metal ball: mass 500 g, volume 50 cm^3 . Calculate density of each and state which is denser.	3	OR	500 mL of cooking oil has mass 450 g. Find its density in g/mL . Then find the volume of 1.8 kg of this oil.	3
(iii)	Explain the displacement method for finding density of an irregular solid. List the steps in order and write the formula used.	3	OR	A rock of mass 80.52 g is lowered into a measuring cylinder; water level rises from 18 mL to 46 mL. Find the volume and density of the rock.	3
(iv)	Compare solids, liquids and gases with reference to: (a) particle spacing, (b) type of	3	OR	Why do liquids take the shape of their container but solids do not? Use the concept	3

	particle motion, (c) relative density.			of attractive forces between particles.	
(v)	Define plasma . State the conditions for its formation from a gas. Give two examples of plasma from daily life.	3	OR	Why is plasma called the fourth state of matter? State two properties that distinguish plasma from an ordinary gas.	3
(vi)	State the relationship between particle motion and temperature . What happens to particle speed and K.E. when a substance is cooled?	3	OR	Define absolute zero and give its value on the Celsius scale. Explain why no further cooling is possible below this point.	3
(vii)	Define internal energy . State the two forms of energy that make it up and explain how heating a substance increases its internal energy.	3	OR	Name three forms of kinetic energy possessed by particles of a substance. In which state of matter is internal energy entirely translational K.E.?	3
(viii)	Name the three temperature scales . State the lower and upper fixed points on each and the number of divisions between them.	3	OR	Convert: (a) 25 °C to Kelvin, (b) 350 K to Celsius, (c) 100 °C to Fahrenheit.	3
(ix)	Define sensitivity of a thermometer. Explain how capillary tube diameter affects sensitivity of a liquid-in-glass thermometer.	3	OR	Define range of a thermometer. A clinical mercury thermometer has range 35 °C to 42 °C. Express this range in Fahrenheit.	3
(x)	Describe the structure and working of a thermocouple thermometer . Name the two junctions and state what quantity is measured to find temperature.	3	OR	Explain why a liquid crystal strip thermometer changes colour with temperature. Give two practical applications of this type.	3
(xi)	What is a thermometric property ? Name the thermometric property used in: (a) liquid-in-glass, (b) constant-volume gas thermometer, (c) resistance thermometer.	3	OR	Explain how bulb size affects the range and how the nature of liquid affects the sensitivity of a liquid-in-glass thermometer.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Q.3: Attempt **all** questions. Answer *either* the main question *or* the OR alternative.

Q.	Question	Marks	OR	OR Question	Marks
3(i)	Define density ($\rho = m/V$). Using particle spacing, explain why solids > liquids > gases in density. A brick (21.6 cm × 10.2 cm × 6.35 cm, mass 3.30 kg) has what density in kg/m ³ ?	5	OR	Describe the displacement method (all six steps, formula, labelled diagram). Solve: an irregular stone of mass 200 g raises water from 40 mL to 73 mL. Find its volume and density.	5
3(ii)	Describe all four states of matter . Compare solid, liquid, gas in a table (arrangement, motion, density). Explain how plasma forms from a gas and give three examples in nature or daily life.	5	OR	Explain the relationship between temperature, particle motion, and internal energy . Define absolute zero (Celsius and Kelvin values). Describe the K.E.-vs-Kelvin graph and identify where absolute zero falls.	5
3(iii)	State all three temperature scales with conversion formulas (C↔K, C↔F). Solve: (a) 37 °C → K and °F; (b) 212 °F → °C and K; (c) 300 K → °C and °F.	5	OR	Explain calibration of a thermometer . Define lower and upper reference points (values for all three scales). Why must fixed points be reproducible? Describe the calibration procedure using ice and steam points.	5
3(iv)	Compare liquid-in-glass and thermocouple thermometers : thermometric property, structure, working, and applications. Then state three ways the structure of a liquid-in-glass thermometer affects its sensitivity, range, and linearity.	5	OR	Name and describe five types of thermometers , stating the thermometric property of each. For each, give one application where it is preferred and briefly explain why.	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Unit 7 — Density and Temperature

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